

ARM-450 — Summary Report

- What it is: a clean-sheet 6-DOF robot arm, 450 mm overall, designed from scratch to

trace a sinusoid with the pen normal to a surface — and then to serve the MASCOT docking / fuel-refilling work.

- Status: geometry, kinematics and the solved path all verified. Mechanically unproven — nothing has been printed.

- Date: 2026-08-21

1. Where the project stands

Overall length	450.0 mm — requirement met exactly
Horizontal reach	360 mm
Mass	1365 g against a 1450 g hard ceiling (85 g headroom)
Degrees of freedom	6, all with real hardware
Reachable workspace	190 litres, dead zone none
Workable (pen held normal, vertical board)	20 % — a patch ~95 × 140 mm
Bearings	6 × 6806 (J1-J3) + 6 × 6706 (J4-J6)
Printed parts	19 pieces, 624 g
Pre-print gate	60 passed, 0 warnings, 0 failures
Geometry audit (STEP B-rep)	52 features verified, 0 outstanding
Verification rounds	5 of 5 clean
Sine tracing	240/240 waypoints, 0.014 mm rms, 0 collisions

The arm does what it was designed to do. The kinematics are verified to the nanometre against the URDF, and Rsine runs on it unmodified.

What changed on 2026-08-21

Three things that would each have wasted a print run:

1. Three bearing seats did not exist. The link halves and the base had no Ø42 pocket at all — each had been cut into a region an earlier cut already emptied, which CadQuery performs silently. The pre-print gate passed 55/55 on them because it checked the parameter, never the geometry. It now reads the STEP B-rep.
2. The sine path was not collision-free. 72 of 240 waypoints had the upper arm grazing the base. Raising the trace 20 mm cleared every pose and improved accuracy 14×.
3. The mass model was wrong. A 0.55 sparse-infill factor was being applied to hollow-modelled parts, discounting the hollowing twice and hiding ~240 g.

Also: the servo horn had nothing to drive — a horn adapter now closes that torque path — and the wrist moved to a smaller bearing, saving 93 g.

2. What was built

Twelve part types, all STEP + STL, all sharing one joint interface — Ø42 pocket, 6806 bearing, Ø30 shaft — so a fit validated on one joint is valid on all six.

base · turret_j1 · link_half_tongue ×2 · link_half_groove ×2 · joint_shaft · shaft_clamp · servo_collar · wrist_j4_housing · wrist_j5_yoke · wrist_j6_output

All six joints take the same Waveshare ST3215. One part number for the whole machine.

Documents: ARM450_REPORT.pdf (analysis) · ARM450_KINEMATICS.pdf (DH, FK, Jacobian, workspace) · ARM450_DRAWINGS.pdf (parts) · ARM450_LESSONS.pdf (mistakes) · ARM450_LOG.pdf (running log) · this summary.

3. The findings that changed the design

The bearing was the whole problem. The original single thrust washer has zero moment capacity — 0.2 mm of clearance became 28.9 mm of wobble at the tool. A preloaded pair at 22 mm spacing brings that to 0.65 mm, a 44× improvement. Preload is the entire story: without it the pair only buys 4×.

Mounts were sized for the wrong load. They were sized for the 2.08 N·m the arm carries, when a servo can deliver 2.94 N·m of stall torque into its own mount during a collision. Whenever an actuator can drive its structure to failure, the actuator's limit is the design load.

Equal links eliminate the dead zone. $|L2 - L3|$ is the inner radius of the reachable annulus. The user's original links were equal; a later revision broke that and opened a Ø194 mm hole.

Stress was never the constraint. Safety factors run 20–400 across every part. The section size is set by the servo that has to fit inside it, not by any load.

4. What is verified, and what is not

Verified numerically:

- DH table vs URDF: 0.0000 µm over 5,000 poses
- Jacobian vs numerical differentiation: 1.85×10^{-8} m/rad
- Chain totals 450 mm, asserted at build time
- Workspace, dead zone, task workspace: 250k-pose sweeps
- Interference across joint limits: 16 % of poses collide, and every one is arm-vs-base, none arm-vs-arm
- FEA on all parts: SF 20–400
- URDF extracted from the CAD with real meshes and inertias

Not verified — and this is the honest gap:

- The 0.65 mm wobble is calculated, never measured. It is the single most valuable

number to obtain.

- Four of six joint parts (J1, J4, J5, J6) are first-pass: no shaft fit check, no servo horn clearance, no cable routing.
- Servo mounting-hole and horn bolt patterns are placeholders. The user's ST3215 STL is an outer shell with no holes modelled.
- Horizontal-table tracing does not work and needs a wrist rethink.

5. Against the guide's brief

Sine motion — achievable now. Vertical board at 240–300 mm, 0.028 mm rms.

MASCOT docking — yes, but by geometry rather than by accuracy. Absolute accuracy is backlash-limited at roughly 5 mm and no structural change fixes that; it would take joint-side encoders. So the docking tool does not ask the arm to be accurate. Its capture cone is a Ø34 mouth over a Ø16 throat, and the measured envelope — probe mesh swept sideways against target mesh until they foul — is ± 9.75 mm, about 2× the arm's backlash-limited error and 7× its repeatable error. The cone converts a positioning problem into a geometry problem, which is the standard answer in docking hardware and the reason the mouth is as oversized as it is.

The latch needs no seventh actuator either: it is a bayonet and J6 turns it.

"Space grade with minor adjustment" — only in the geometric sense. The kinematics, joint architecture and fastening scheme all survive a material swap. PLA does not (outgassing, 60 °C glass transition), hobby servos do not (no vacuum lubrication), and wet-grease bearings do not. Re-made in Al 6061 the same design is 1.13 kg and about 10× stiffer. Worth agreeing with the guide which meaning is intended before promising it.

6. Next three steps

1. Measure the wobble. Assemble one joint, dial indicator at 360 mm, push both ways, record the hysteresis before and after preload. That single before/after pair tests the entire bearing theory.
2. Measure a real ST3215 and correct four numbers in cad/params.py. Every mounting hole regenerates.
3. Reprint the link halves at 119 mm. They were shortened from 145 mm to buy wrist length; reach is unchanged.